

# Robust Reinforcement Learning for Autonomous Driving Under Observation Corruption and Partial Observability

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**Abstract**—Autonomous driving agents must remain reliable when sensor observations are noisy, incomplete, or partially unavailable. Many reinforcement learning evaluations assume clean and complete observations, which is unrealistic for real-world driving where sensors may suffer from noise, occlusion, and perception failure. This paper presents a robustness evaluation of Proximal Policy Optimization (PPO) and Recurrent PPO for autonomous driving in the *highway-env* simulator. We formulate the problem as reinforcement learning under observation corruption and partial observability. The study evaluates clean driving, Gaussian observation noise, reward shaping, vehicle occlusion, extreme occlusion, and long-training validation. Results show that PPO remains stable under Gaussian noise, while vehicle occlusion is more challenging because it removes critical environmental information. Recurrent PPO was evaluated as a memory-based solution, but it did not consistently outperform standard PPO under the current training budget. The results indicate that missing observations are more harmful than additive sensor noise and that robust autonomous driving policies should be evaluated using both performance and safety metrics.

**Index Terms**—Reinforcement Learning, PPO, Recurrent PPO, Autonomous Driving, Robustness, Partial Observability, Sensor Noise, Highway-Env

## I. INTRODUCTION

Reinforcement Learning (RL) has become an important framework for sequential decision-making problems such as robotics, control, and autonomous driving. In autonomous driving, an agent must learn how to select actions such as acceleration, braking, and lane changes based on observations of the surrounding traffic environment. Deep RL methods allow agents to learn such behaviors through interaction rather than manually designed rules.

However, most RL evaluations assume that the agent receives complete and clean observations. This assumption is not realistic for autonomous driving. Real-world sensors such as cameras, LiDAR, radar, and GPS can be affected by noise, occlusion, weather, latency, and perception errors. A policy that performs well in a clean simulator may fail when the observation space becomes corrupted.

This project investigates the robustness of PPO and Recurrent PPO under sensor noise and partial observability. Instead of only training a policy in a standard environment, we evaluate how policy performance changes when observations are corrupted by Gaussian noise or vehicle occlusion. We also

evaluate whether an LSTM-based recurrent policy can improve robustness under missing observations.

The main contributions are:

- We formulate noisy and occluded autonomous driving as an RL problem under observation corruption.
- We evaluate PPO and Recurrent PPO under clean, noisy, occluded, and extreme occlusion settings.
- We conduct noise sensitivity, occlusion sensitivity, and 100k-step validation experiments.
- We analyze robustness using reward, collision rate, and average speed rather than reward alone.

## II. RELATED WORK

### A. Reinforcement Learning for Autonomous Driving

RL has been widely studied for autonomous driving because driving can naturally be formulated as a sequential decision-making problem. Simulation environments such as *highway-env* provide lightweight platforms for testing lane changing, overtaking, and traffic interaction policies. These environments are useful for studying algorithmic behavior before moving to high-fidelity simulators.

### B. Proximal Policy Optimization

PPO is a policy-gradient algorithm designed to improve training stability by limiting large policy updates. PPO has become popular due to its simplicity, robustness, and strong empirical performance across many control tasks. Its clipped objective prevents the new policy from moving too far away from the previous policy during optimization.

### C. Partial Observability and Recurrent Policies

Autonomous driving is often partially observable because the agent may not have complete information about all surrounding vehicles. Recurrent neural networks, especially LSTM models, are commonly used to maintain memory over time. Recurrent PPO extends PPO by adding a hidden state that can summarize previous observations, which may help when the current observation is incomplete.

### III. PROBLEM FORMULATION

#### A. Markov Decision Process

A standard RL task can be formulated as a Markov Decision Process (MDP):

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma), \quad (1)$$

where  $\mathcal{S}$  is the state space,  $\mathcal{A}$  is the action space,  $P(s_{t+1}|s_t, a_t)$  is the transition function,  $R(s_t, a_t)$  is the reward function, and  $\gamma$  is the discount factor.

The objective is to learn a policy  $\pi_\theta$  that maximizes expected discounted return:

$$J(\pi_\theta) = \mathbb{E}_{\pi_\theta} \left[ \sum_{t=0}^T \gamma^t r_t \right]. \quad (2)$$

#### B. Observation Corruption

In realistic driving, the agent does not always receive the clean observation  $o_t$ . Instead, it receives a corrupted observation:

$$\tilde{o}_t = f(o_t, c), \quad (3)$$

where  $c$  represents the corruption condition, such as Gaussian noise or vehicle occlusion.

The policy therefore acts according to:

$$a_t \sim \pi_\theta(a_t|\tilde{o}_t). \quad (4)$$

#### C. POMDP Interpretation

When observations are corrupted or incomplete, the task becomes partially observable. The underlying state  $s_t$  is not fully known, and the policy must act using  $\tilde{o}_t$ . This can be viewed as a Partially Observable Markov Decision Process (POMDP):

$$\mathcal{P} = (\mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma), \quad (5)$$

where  $\Omega$  is the observation space and  $O(o_t|s_t)$  is the observation function.

#### D. Robust RL Objective

We define a robustness objective over a set of corruption conditions  $\mathcal{C}$ :

$$J_{\text{robust}}(\pi) = \mathbb{E}_{c \sim \mathcal{C}} [J(\pi, c)]. \quad (6)$$

A robust policy should maintain high return and low collision rate across multiple corruption settings.

### IV. METHODOLOGY

#### A. Environment

The experiments use the *highway-env* simulator. The agent controls an ego vehicle in a highway traffic environment and must learn lane-changing and driving behavior.



Fig. 1. Highway-env autonomous driving environment used in the experiments.

#### B. PPO

PPO optimizes a clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (7)$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}. \quad (8)$$

Here,  $\hat{A}_t$  is the advantage estimate and  $\epsilon$  controls the clipping range.

#### C. Recurrent PPO

Recurrent PPO introduces a hidden memory state:

$$h_t = g_\theta(h_{t-1}, \tilde{o}_t), \quad (9)$$

and selects actions using:

$$a_t \sim \pi_\theta(a_t|\tilde{o}_t, h_{t-1}). \quad (10)$$

This allows the policy to use previous observations when the current observation is incomplete.

#### D. Gaussian Noise Model

Gaussian observation noise is modeled as:

$$\tilde{o}_t = o_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2 I). \quad (11)$$

Here,  $\sigma$  controls the severity of the observation noise.

#### E. Vehicle Occlusion Model

Vehicle occlusion is modeled using a binary mask:

$$\tilde{o}_t = m_t \odot o_t, \quad (12)$$

where  $m_t$  is a binary mask and  $\odot$  denotes element-wise multiplication. Unlike Gaussian noise, occlusion removes information from the observation space.

#### F. Reward Shaping

Eco-driving reward shaping is introduced as:

$$r'_t = r_t - \alpha |a_t^{\text{acc}}| - \beta |a_t^{\text{steer}}|, \quad (13)$$

where  $a_t^{\text{acc}}$  and  $a_t^{\text{steer}}$  represent acceleration and steering components.

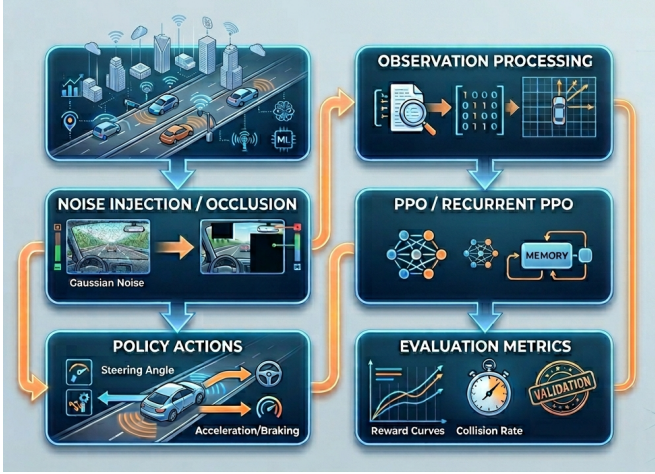


Fig. 2. Experimental pipeline for robustness evaluation.

## V. EXPERIMENTAL DESIGN

The experiments evaluate PPO and Recurrent PPO under the following settings:

- PPO Clean Baseline
- PPO Gaussian Noise
- PPO Eco + Noise
- PPO Strong Noise
- PPO Extreme Occlusion
- Recurrent PPO Extreme Occlusion
- PPO Extreme Occlusion 100k Validation

The evaluation metrics are mean reward, standard deviation of reward, collision rate, and average speed.

Mean reward is:

$$\bar{R} = \frac{1}{N} \sum_{i=1}^N R_i. \quad (14)$$

Collision rate is:

$$C = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{collision}_i]. \quad (15)$$

Average speed is:

$$\bar{v} = \frac{1}{T} \sum_{t=1}^T v_t. \quad (16)$$

## VI. RESULTS AND ANALYSIS

### A. Main Comparison

Table I summarizes the final experimental results.

The results show that PPO remains stable across most settings. Gaussian noise did not significantly degrade performance. The Recurrent PPO model under extreme occlusion produced a high collision rate, suggesting that recurrence alone does not guarantee robustness under severe partial observability.

TABLE I  
FINAL COMPARISON OF PPO AND RECURRENT PPO UNDER DIFFERENT CONDITIONS.

| Experiment                 | Reward | Collision | Speed |
|----------------------------|--------|-----------|-------|
| PPO Clean 10k              | 12.75  | 0.10      | 19.99 |
| PPO Gaussian Noise 10k     | 13.84  | 0.00      | 20.04 |
| PPO Eco + Noise 10k        | 14.11  | 0.00      | 20.04 |
| PPO Strong Noise 10k       | 14.06  | 0.00      | 20.04 |
| PPO Extreme Occlusion 20k  | 14.06  | 0.00      | 20.04 |
| Recurrent PPO Extreme 20k  | 12.36  | 0.70      | 23.71 |
| PPO Extreme Occlusion 100k | 14.00  | 0.05      | 20.15 |

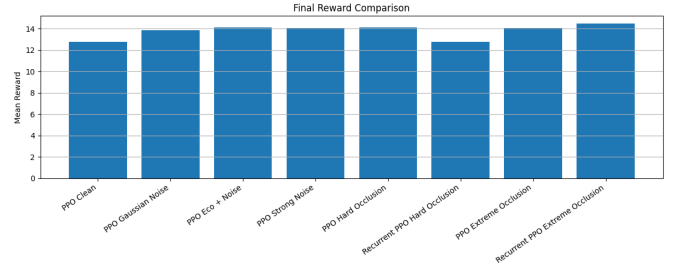


Fig. 3. Mean reward comparison across experimental settings.

### B. Noise Sensitivity

The noise sensitivity study evaluates PPO across increasing Gaussian noise levels.

The results show that PPO remains stable across Gaussian noise levels. This supports the hypothesis that additive noise perturbs observations while preserving useful information.

### C. Occlusion Sensitivity

The occlusion sensitivity study evaluates PPO under increasing vehicle occlusion probability.

Occlusion is more difficult than Gaussian noise because it removes information instead of perturbing it. This increases uncertainty and makes the environment more strongly partially observable.

### D. Long-Training Validation

The PPO extreme occlusion model was also trained for 100k steps. The result achieved mean reward 13.999, collision rate 0.05, and average speed 20.154. This confirms that PPO remains stable under longer training in the extreme occlusion setting.

## VII. ABLATION STUDY

### A. Effect of Gaussian Noise

The noise ablation shows that PPO maintains stable rewards across increasing noise. The key reason is that Gaussian noise does not fully remove environmental structure. The agent still receives information about nearby vehicles, lane positions, and speed.

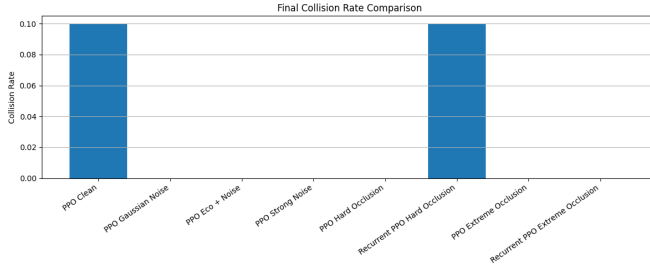


Fig. 4. Collision rate comparison across experimental settings.

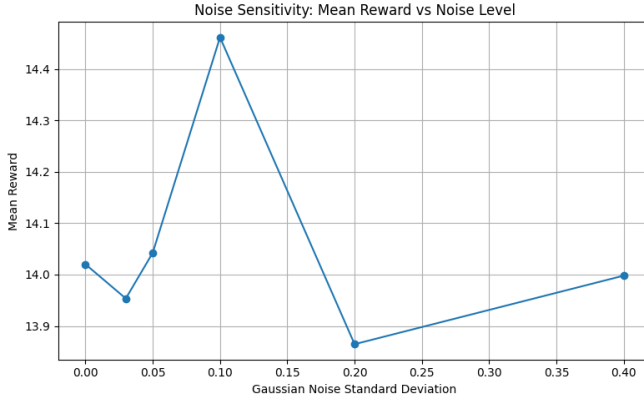


Fig. 5. Mean reward under increasing Gaussian observation noise.

### B. Effect of Vehicle Occlusion

Occlusion has a stronger impact because it removes vehicle observations from the input. This creates hidden state uncertainty and makes the policy depend on incomplete information. The occlusion experiment therefore represents a stronger POMDP than Gaussian noise.

### C. Effect of Recurrent Memory

Recurrent PPO was expected to help under partial observability because LSTM memory can use past observations. However, the experiment showed that Recurrent PPO did not consistently outperform standard PPO. This suggests that memory-based policies require longer training, better tuning, or curriculum learning to fully exploit temporal information.

## VIII. DISCUSSION

The experimental results support the distinction between information perturbation and information removal. Gaussian noise perturbs the observation but preserves most of the useful structure. In contrast, occlusion removes important information and increases state uncertainty.

The results also show that reward alone is not sufficient to evaluate safety. Recurrent PPO under extreme occlusion had high collision rate despite being a memory-based model. Therefore, collision rate and average speed must be included as safety-oriented evaluation metrics.

The 100k-step PPO validation indicates that standard PPO can learn stable behavior under difficult observation conditions. However, the results do not imply that PPO is universally

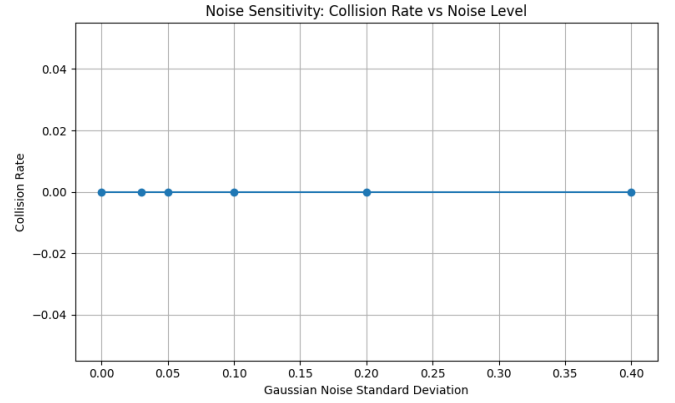


Fig. 6. Collision rate under increasing Gaussian observation noise.

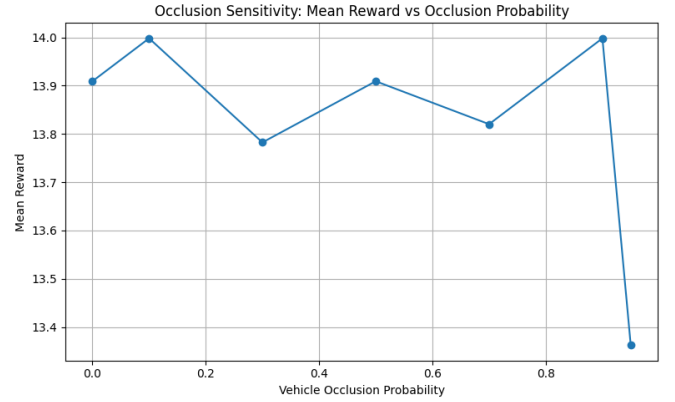


Fig. 7. Mean reward under increasing vehicle occlusion probability.

robust. They show robustness within the tested highway-env settings.

## IX. LIMITATIONS

This study has several limitations. First, highway-env is a simplified driving simulator and does not fully represent real-world traffic complexity. Second, the occlusion model uses a simplified masking strategy. Third, Recurrent PPO was trained with limited timesteps, which may not be sufficient for fully learning memory-based behavior. Finally, the experiments were conducted with limited random seeds due to compute constraints.

## X. FUTURE WORK

Future work can extend this project by training Recurrent PPO for longer timesteps, applying curriculum-based noise and occlusion schedules, evaluating denser traffic scenarios, and testing in more realistic simulators such as CARLA. Additional safety metrics such as time-to-collision, lane-change smoothness, and jerk can also improve evaluation.

## XI. CONCLUSION

This paper presented a research-style robustness evaluation of PPO and Recurrent PPO for autonomous driving under

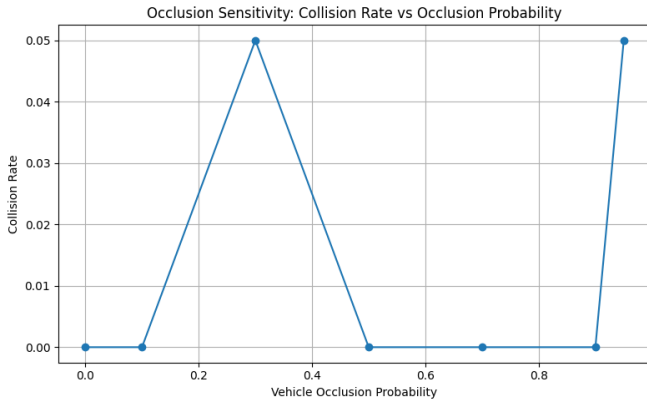


Fig. 8. Collision rate under increasing vehicle occlusion probability.

observation corruption and partial observability. The results show that PPO remains robust under Gaussian noise, while vehicle occlusion represents a more challenging setting because it removes critical information from the observation space. Recurrent PPO did not consistently outperform PPO under the current training budget, indicating that memory alone is not enough without sufficient training and tuning. Overall, the study demonstrates that robust autonomous driving policies must be evaluated using both performance and safety metrics.

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